

Lecture 29

Singular Value Decomposition = SVD

$$A = Q \Lambda Q^T$$

↓
positive symmetric definite

$$A = U \Sigma V^T$$

↑
diagonal

orthogonal

Suppose we have a vector v (in row space of A)
and $u = Av$ in col space of A

* The we say that v is the orthogonal basis of row space of A . V

* $u \rightarrow$ orthogonal basis of $\text{col}(A)$

* $\sigma \rightarrow$ multiplying factors for mapping orthogonal basis

$$A \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_r \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & & & | \\ u_1 & \dots & u_r \\ | & & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \end{bmatrix}$$

$$AV = U\Sigma$$

eg $\begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$

$$v_1, v_2 \rightarrow \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$u_1, u_2 \rightarrow \begin{bmatrix} 4 \\ -3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ maybe?}$$

→ Basis for null space goes in V & U after the respective bases. and in Σ as zeroes on diagonal.

Fact $A = U \Sigma V^T$

$$A^T A = V \Sigma^T \underbrace{U^T U}_{\mathbf{I}} \Sigma V^T$$

$$\boxed{A^T A = V \Sigma^2 V^T}$$

So, to find the V & U ,
we can find eigenvectors of
 $A^T A$ & AA^T respectively.

eg $A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$ $A^T = \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 25 & 7 \\ 7 & 25 \end{bmatrix}$$

$$(25 - \lambda)^2 - 49 = 0$$

$$(\lambda - 25 - 7)(\lambda - 25 + 7) = 0$$

$$\lambda = 32, 18$$

$$\Rightarrow V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\Sigma^2 = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

$$\Sigma = \sqrt{2} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$$

→ To find U

$$A A^T = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

$$\lambda = 32, 18$$

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

#

The order of λ 's should stay the same!!

$$A = U \Sigma V^T$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \right) \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ 3 & -3 \end{bmatrix}$$

Example 2

$$\begin{bmatrix} 4 & 8 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} 80 & 60 \\ 60 & 48 \end{bmatrix}$$

$$\lambda = 0, 128$$

$$A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$$

$$C(A^T) = \lambda \begin{bmatrix} 4 & 3 \end{bmatrix}$$

$$N(A^T) = \lambda \begin{bmatrix} 3 & -4 \end{bmatrix}$$

$$C(A) = \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$N(A) = \lambda \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ 2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix}}_U \underbrace{\begin{bmatrix} 5\sqrt{5} & 0 \\ 0 & 0 \end{bmatrix}}_{\Lambda} \underbrace{\begin{bmatrix} 4/5 & 3/5 \\ 3/5 & -4/5 \end{bmatrix}}_{V^T}$$

Basis for $C(A)$
Basis for $N(A)$

Basis for $C(A^T)$
Basis for $N(A^T)$

$$\frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \sqrt{5} \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$$

$$\# \quad A v_i = \sigma_i u_i$$

$v_1 \dots v_r$	$\rightarrow$	orthonormal basis	for	$C(A^T)$
$v_{r+1} \dots v_n$	$\rightarrow$	"	"	$N(A^T)$
$u_1 \dots u_r$	$\rightarrow$	"	"	$C(A)$
$u_{r+1} \dots u_m$	$\rightarrow$	"	"	$N(A)$